SAP ABAP Projekt

CDS Views und Join-Abfragen (Flight Scenario)

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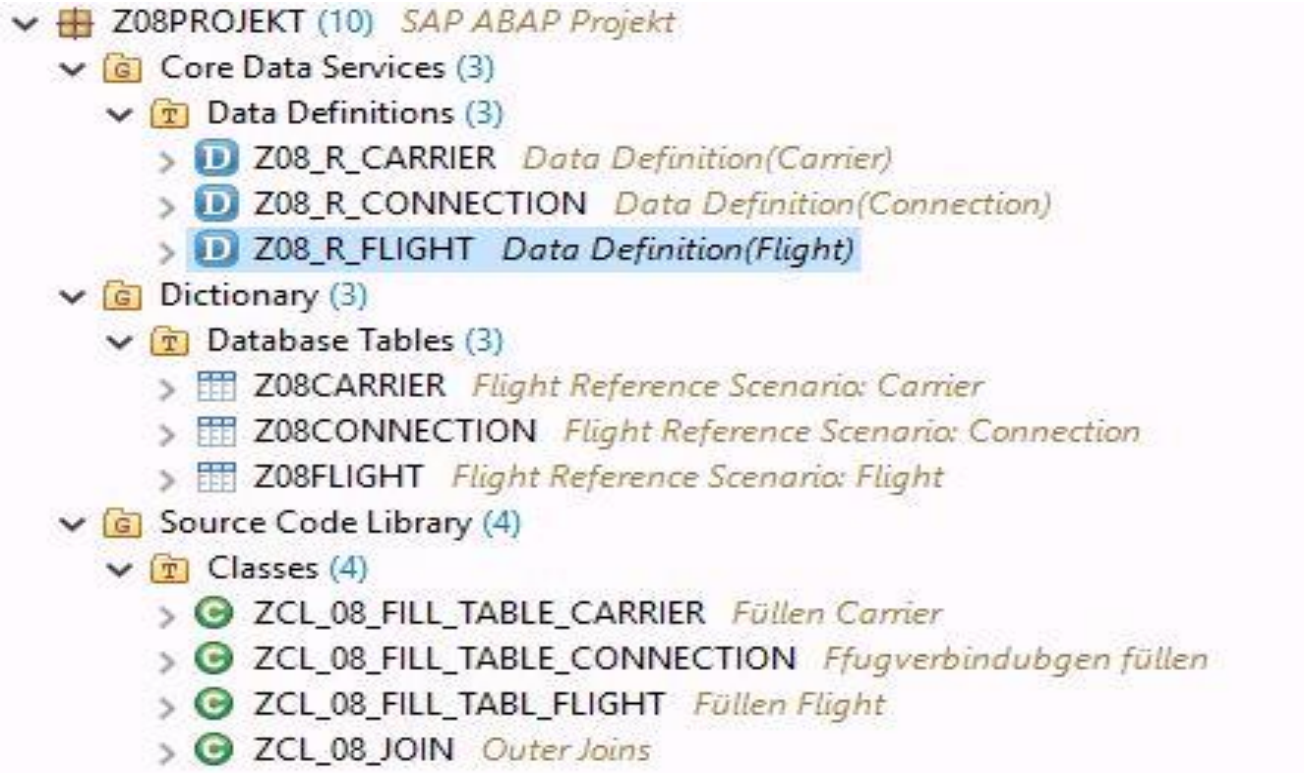
Projektübersicht:

In diesem Projekt werden Flugdaten (Carrier, Connection, Flight) mithilfe von CDS Views und Open SQL Join-Abfragen modelliert und ausgewertet.



Verwendete Datenbanktabellen

- Z08CARRIER – Fluggesellschaften
- Z08CONNECTION – Flugverbindungen
- Z08FLIGHT – Flüge

Befüllen der Carrier-Tabelle

ZCL_08_FILL_TABLE_CARRIER ▸ IF_OO_ADT_CLASSRUN~MAIN ▸ loop

```
CLASS zcl_08_fill_table_carrier DEFINITION
  PUBLIC
  FINAL
  CREATE PUBLIC .

  PUBLIC SECTION.

    INTERFACES if_oo_adt_classrun .
  PROTECTED SECTION.
  PRIVATE SECTION.
ENDCLASS.



CLASS zcl_08_fill_table_carrier IMPLEMENTATION.


  METHOD if_oo_adt_classrun~main.

    TYPES: BEGIN OF ts_strings,
             client        TYPE string,
             carrier_id    TYPE string,
             name          TYPE string,
             currency_code TYPE string,
           END OF   ts_strings.

      DATA: ls_strings TYPE ts_strings.

      TYPES: ts_string TYPE string.

      DATA: lt_strings TYPE TABLE OF ts_string.
      DATA: lt_source4db  TYPE TABLE OF Z08carrier.
      DATA: ls_source4db  LIKE LINE OF lt_source4db.


  lt_strings = VALUE #(
                        ( |100#AA#American Airlline#USD| )
                        ( |100#AC#Air Canada#USD| )
                        ( |100#AF#Air France#EUR| )
                        ( |100#AZ#Alitalia Societa Aerea Italiana S.p.A.#EUR| )
                        ( |100#BA#British Airways p.l.c.#GBP| )
```

ZCL_08_FILL_TABLE_CARRIER ▸ IF_OO_ADT_CLASSRUN~MAIN ▸ loop

```
  lt_strings = VALUE #(
                          ( |100#AA#American Airlline#USD| )
                          ( |100#AC#Air Canada#USD| )
                          ( |100#AF#Air France#EUR| )
                          ( |100#AZ#Alitalia Societa Aerea Italiana S.p.A.#EUR| )
                          ( |100#BA#British Airways p.l.c.#GBP| )
                          ( |100#FJ#Air Pacific Limited t/a Fiji Airway#USD| )
                          ( |100#CO#Cobaltair Ltd. dba Cobalt#USD| )
                          ( |100#LH#Deutsche Lufthansa AG#EUR| )
                          ( |100#NG#AL-Naser Wings#EUR| )
                          ( |100#JL#Japan Airlines Co. Ltd.#JPY| )
                    ).



       LOOP AT lt_strings INTO DATA(string_line).
         SPLIT string_line AT '#'
             INTO ls_strings-client
                  ls_strings-carrier_id
                  ls_strings-name
                  ls_strings-currency_code.

           "Move data to DB structure
                 ls_source4db = CORRESPONDING #( ls_strings ).

            "Add to internal DB table
                 APPEND ls_source4db TO lt_source4db.

      ENDLOOP.


       MODIFY z08carrier FROM TABLE @lt_source4db.
       COMMIT WORK.
       out->write( |DB-Table modified?: SY-SUBRC = { sy-subrc } SY-DBCNT = { sy-dbcnt }| ).



  ENDMETHOD.
ENDCLASS.
```

Befüllen der Connection-Tabelle

ZCL_08_FILL_TABLE_CONNECTION

```
CLASS zcl_08_fill_table_connection DEFINITION
  PUBLIC
  FINAL
  CREATE PUBLIC .

  PUBLIC SECTION.

    INTERFACES if_oo_adt_classrun .
  PROTECTED SECTION.
  PRIVATE SECTION.
ENDCLASS.



CLASS zcl_08_fill_table_connection IMPLEMENTATION.


  METHOD if_oo_adt_classrun~main.

  TYPES: BEGIN OF ts_strings,
              client          TYPE string,
              carrier_id      TYPE string,
              connection            TYPE string,
              airport_from_id TYPE string,
             airport_to_id    TYPE string,
             departure_time   TYPE string,
             arrival_time     TYPE string,
             distance         TYPE string,
             distance_unit    TYPE string,
          END OF   ts_strings.

      DATA: ls_strings TYPE ts_strings.



      TYPES: ts_string TYPE string.

      DATA: lt_strings TYPE TABLE OF ts_string.
```

ZCL_08_FILL_TABLE_CONNECTION

```
  lt_strings = VALUE #(
                            ( |100#AA#0015#JFK#SFO#071300#100400#4156#KM| )
                            ( |100#AA#2678#HAV#MIA#061500#103000#520#KM| )
                            ( |100#AA#0017#MIA#HAV#071900#080300#520#KM| )
                            ( |100#AA#0018#SFO#JFK#064000#150600#4156#KM| )
                            ( |100#AC#0322#MIA#EWR#201700#231900#1752#KM| )
                            ( |100#AZ#0788#VCE#NRT#132500#101300#9595#KM| )
                            ( |100#AZ#0789#NRT#VCE#142600#213100#9595#KM| )
                            ( |100#JL#0407#NRT#FRA#132300#155600#9379#KM| )
                            ( |100#FJ#0408#FRA#NRT#202500#154000#9379#KM| )
                            ( |100#LH#0400#FRA#JFK#101000#113400#6162#KM| )
                         ).


      LOOP AT lt_strings INTO DATA(string_line).
        SPLIT string_line AT '#'
            INTO ls_strings-client
                 ls_strings-carrier_id
                 ls_strings-connection
                 ls_strings-airport_from_id
                 ls_strings-airport_to_id
                 ls_strings-departure_time
                 ls_strings-arrival_time
                 ls_strings-distance
                 ls_strings-distance_unit.

           "Move data to DB structure
                ls_source4db = CORRESPONDING #( ls_strings ).

            "Add to internal DB table
                APPEND ls_source4db TO lt_source4db.

     ENDLOOP.


      MODIFY z08connection FROM TABLE @lt_source4db.
      COMMIT WORK.
      out->write( |DB-Table modified?: SY-SUBRC = { sy-subrc } SY-DBCNT = { sy-dbcnt }| ).

  ENDMETHOD.
ENDCLASS.
```

Befüllen der Flight-Tabelle

ZCL_08_FILL_TABL_FLIGHT ▸ IF_OO_ADT_CLASSRUN~MAIN

```
  lt_strings = VALUE #(
                        ( |100#SQ#0015#20250514#10818.00 #SGD#767-200#260#223| )
                        ( |100#SQ#0015#20240718#5950.00 #SGD#A340-600#330#168| )
                        ( |100#SQ#2678#20250515#11765.00 #SGD#747-400#385#350| )
                        ( |100#SQ#0018#20240719#10953.00 #SGD#747-400#385#334| )
                        ( |100#SQ#0322#20250515#2359.00 #SGD#767-200#260#132| )
                        ( |100#SQ#0011#20240719#4880.00 #SGD#A340-600#330#310| )
                        ( |100#SQ#0012#20250517#4665.00 #SGD#767-200#260#236| )
                        ( |100#SQ#0012#20240721#2574.00 #SGD#747-400#385#215| )
                        ( |100#UA#0058#20250512#6629.00 #USD#767-200#260#200| )
                        ( |100#UA#0058#20240716#4996.00 #USD#747-400#385#231| )
                       ).

      LOOP AT lt_strings INTO DATA(string_line).
        SPLIT string_line AT '#'
            INTO ls_strings-client
                 ls_strings-carrier_id
                 ls_strings-connection_id
                 ls_strings-flight_date
                 ls_strings-price
                 ls_strings-currency_code
                 ls_strings-plane_type_id
                 ls_strings-seats_max
                 ls_strings-seats_occupied.

          "Move data to DB structure
               ls_source4db = CORRESPONDING #( ls_strings ).

           "Add to internal DB table
               APPEND ls_source4db TO lt_source4db.

     ENDLOOP.

      MODIFY z08flight FROM TABLE @lt_source4db.
      COMMIT WORK.
      out->write( |DB-Table modified?: SY-SUBRC = { sy-subrc } SY-DBCNT = { sy-dbcnt }| ).



  ENDMETHOD.
ENDCLASS.
```

JOINS Mit Datenausgabe

ZCL_08_JOIN ▸ IF_OO_ADT_CLASSRUN~MAIN

```
CLASS zcl_08_join DEFINITION
  PUBLIC
  FINAL
  CREATE PUBLIC .

  PUBLIC SECTION.

    INTERFACES if_oo_adt_classrun .
  PROTECTED SECTION.
  PRIVATE SECTION.
ENDCLASS.

CLASS zcl_08_join IMPLEMENTATION.

  METHOD if_oo_adt_classrun~main.

 SELECT
  FROM z08carrier AS a
  LEFT OUTER JOIN z09flight AS c
    ON a~carrier_id = c~carrier_id
  FIELDS
    a~carrier_id,
    a~name,
    a~currency_code
  WHERE a~currency_code = 'USD'
  INTO TABLE @DATA(lt_result).
    out->write( lt_result ).

 SELECT
     FROM z08carrier
     INNER JOIN z08connection
      ON z08carrier~carrier_id = z08connection~carrier_id
     FIELDS
        z08carrier~client,
        z08carrier~carrier_id,
        z08carrier~name,
        z08carrier~currency_code
     INTO TABLE @DATA(lt_resultl).
    out->write( lt_resultl ).
  ENDMETHOD.
ENDCLASS.
```

CDS Views – Datenmodellierung

CDS View: Flight – Definition und Datenausgabe

[A2J] Z08_R_FLIGHT

```
1  @AbapCatalog.viewEnhancementCategory: [#NONE]
2  @AccessControl.authorizationCheck: #NOT_REQUIRED
3  @EndUserText.label: 'Data Definition(Flight)'
4  @Metadata.ignorePropagatedAnnotations: true
5  @ObjectModel.usageType:{
6      serviceQuality: #X,
7      sizeCategory: #S,
8      dataClass: #MIXED
9  }
10 define view entity Z08_R_FLIGHT
11      as select from z08flight
12      association [0..*] to Z08_R_CONNECTION as _Connection
13         on $projection.CarrierId = _Connection.CarrierId
14 {
15     key carrier_id as CarrierId,
16     key connection_id as ConnectionId,
17     key flight_date as FlightDate,
18     @Semantics.amount.currencyCode: 'CurrencyCode'
19     price as Price,
20     currency_code as CurrencyCode,
21     plane_type_id as PlaneTypeId,
22     seats_max as SeatsMax,
23     seats_occupied as SeatsOccupied,
24     _Connection
25 }
26
```

[A2J] Z08_R_FLIGHT [A2J] Z08_R_FLIGHT

Z08_R_FLIGHT

Data Preview

find pattern 10 rows retrieved - 4 ms SQL Console

CarrierId	ConnectionId	FlightDate	Price	CurrencyCode	PlaneTypeId	SeatsMax	SeatsOccupied
SQ	0015	2025-05-14	10818.00	SGD	767-200	260	223
SQ	0015	2024-07-18	5950.00	SGD	A340-600	330	168
SQ	2678	2025-05-15	11765.00	SGD	747-400	385	350
SQ	0018	2024-07-19	10953.00	SGD	747-400	385	334
SQ	0322	2025-05-15	2359.00	SGD	767-200	260	132
SQ	0011	2024-07-19	4880.00	SGD	A340-600	330	310
SQ	0012	2025-05-17	4665.00	SGD	767-200	260	236
SQ	0012	2024-07-21	2574.00	SGD	747-400	385	215
UA	0058	2025-05-12	6629.00	USD	767-200	260	200
UA	0058	2024-07-16	4996.00	USD	747-400	385	231

CDS View: Connection Definition und Datenausgabe

[A2J] Z08_R_CONNECTION

```
@AbapCatalog.viewEnhancementCategory: [#NONE]
@AccessControl.authorizationCheck: #NOT_REQUIRED
@EndUserText.label: 'Data Definition(Connection)'
@Metadata.ignorePropagatedAnnotations: true
@ObjectModel.usageType:{
    serviceQuality: #X,
    sizeCategory: #S,
    dataClass: #MIXED
}
define view entity Z08_R_CONNECTION
     as select from z08connection
     association [1..1] to Z08_R_CARRIER as _Carrier
        on $projection.CarrierId = _Carrier.CarrierId
{
    key carrier_id        as CarrierId,
    key connection_id     as ConnectionId,
    airport_from_id       as AirportFromId,
    airport_to_id         as AirportToId,
    departure_time        as DepartureTime,
    arrival_time          as ArrivalTime,
    distance              as Distance,
    distance_unit         as DistanceUnit,
    _Carrier.Name         as Name,
    _Carrier.CurrencyCode as CurrencyCode
}
```

[A2J] Z08_R_CONNECTION | [A2J] Z08_R_CONNECTION

Z08_R_CONNECTION

Data Preview

find pattern | 6 rows retrieved - 4 ms | SQL Console | Number of Entries | Configure... | Show Log | Max. R

CarrierId	ConnectionId	AirportFromId	AirportToId	DepartureTime	ArrivalTime	Distance	DistanceUnit	Name	CurrencyCode
AA	0000	SFO	JFK	06:40:00 AM	03:06:00 PM	4.156	KM	American A...	USD
AC	0000	MIA	EWR	08:17:00 PM	11:19:00 PM	1.752	KM	Air Canada	USD
AZ	0000	NRT	VCE	02:26:00 PM	09:31:00 PM	9.595	KM	Alitalia Soci...	EUR
JL	0000	NRT	FRA	01:23:00 PM	03:56:00 PM	9.379	KM	Japan Airlin...	JPY
FJ	0000	FRA	NRT	08:25:00 PM	03:40:00 PM	9.379	KM	Air Pacific L...	USD
LH	0000	FRA	JFK	10:10:00 AM	11:34:00 AM	6.162	KM	Deutsche L...	EUR